

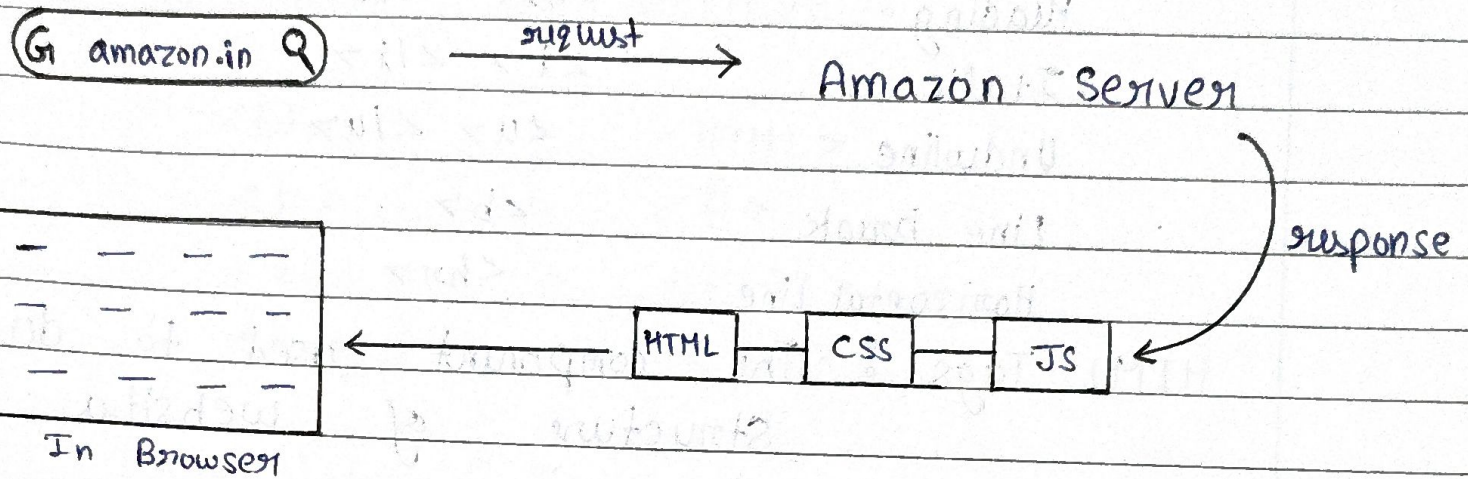
7.

What is Internet?

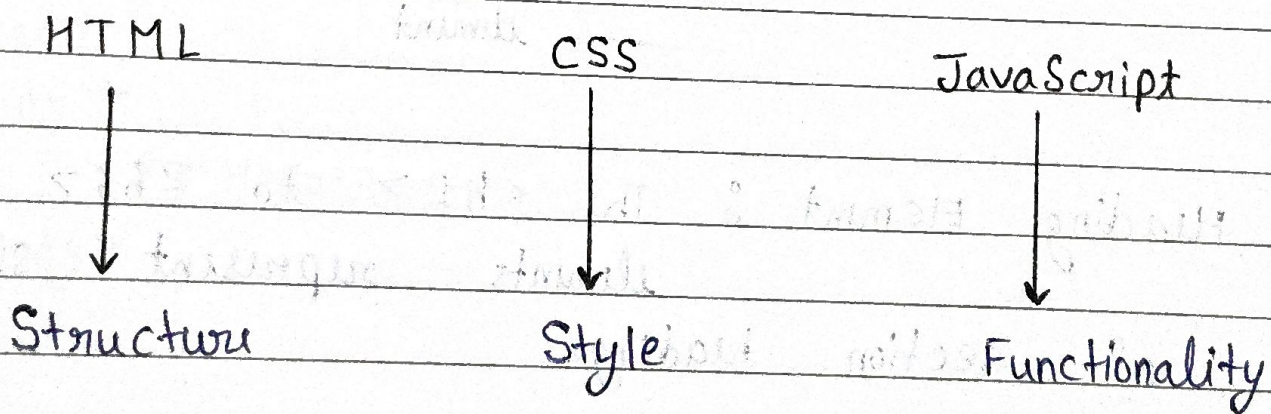
Interconnected network of computers around the world.

What is Web Development?

Building websites for the internet.



जिस वीबसाइट को हम सर्च करते हैं सबसे पहले उसके सर्वर में Request जाती है और उस Code (HTML, CSS, JS) देता है Server Response और उस Code को Browser उसे अपनी website के Display करके दिखा देता है।



HTML: Hyper Text Markup Language

→ Structure and formatting.

HTML Elements : Standard elements that browser recognizes.

Paragraph

`<p>` `</p>`

Bold

`<b>` `</b>`

Heading

`<h1>` `</h1>`

Italic

`<i>` `</i>`

Underline

`<u>` `</u>`

Line break

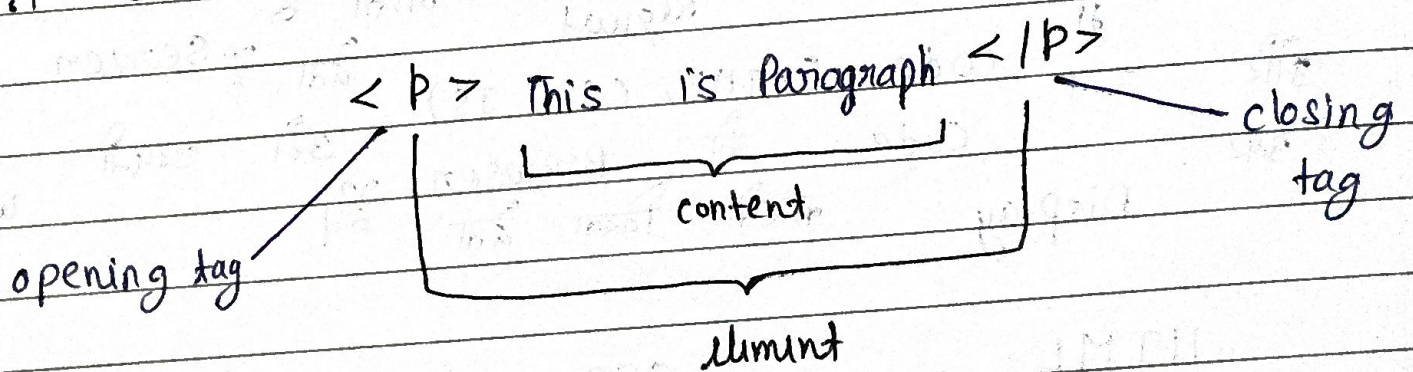
`<br>`

Horizontal Line

`<hr>`

HTML Tags : The component used to design the structure of websites are called HTML tags.

A container for some content or other tags



Heading Element : The `<h1>` to `<h6>` HTML elements represent six levels of section headings.

`<h1>` is the highest section level and `<h6>` is the lowest.

Comments in HTML : This is a part of code that should not be parsed.

`<!-- This is an HTML comment -->`

HTML is not case sensitive :

`<html> = <HTML>`

`<p> = <P>`

`<head> = <HEAD>`

`<body> = <BODY>`

HTML Boilerplate : This is the standard format or skeleton of writing

HTML code.

`<!DOCTYPE html>`

`<html>`

`<head>`

`<title> Page Title </title>`

`</head>`

`<body>`

`<p> Hello World! </p>`

`</body>`

`</html>`

HTML Attributes : Attributes are used to add more information to the tag.

`<html lang = "en">`

attribute

↳ that shows that html was english language.

8. Lists in HTML :

Unordered List -

```
<ul>
```

```
<li> </li>
```

```
<li> </li>
```

```
</ul>
```

Ordered List -

```
<ol> on <ol type = "A">
```

```
<li> </li>
```

```
<li> </li>
```

```
</ol>
```

Anchor Element : Used to add links to your page.

```
<a href = "https://google.com"> Google </a>
```

absolute

↳ Internet or Link

relative

↳ File or Link

href = hyper text reference

↳

This is attribute.

Image Element : Used to add image to your page.

```
<img src = "image.png" alt = "Random Image">
```

source attribute

relative url

(Image Link file/online)

Video Tag in HTML :

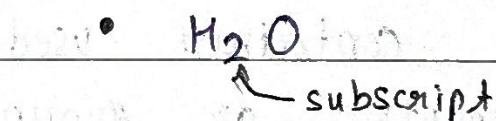
```
<video controls width = "250" >  
<source src = "video.webm" type = "video/mp4" />  
                ↳ video link
```

Note :

Sub & Sup Tag :

• $a^2 = b^2 + c^2$ ← superscript

a < sup > 2 < /sup >



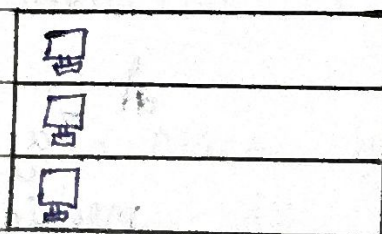
H < sub > 2 < /sub > O

9. Inline Elements : Takes up only necessary width.
Don't start from new line.



Block Element : Takes up the full-width available
(whole block)

Start from new line.



Note :

Heading Element ⇒ Block : Anchor Tag ⇒ Inline

Paragraph Element ⇒ Block : Image Element ⇒ Inline

Div Element :

↳ content division element

Div is a container used to hold other HTML elements or group elements together.

It is a Block element

`<div>` `</div>`

Span Element :

↳ area

Span is also a generic container used to hold other HTML elements or group elements together.

It is an Inline element

`<span>` `</span>`

Semantic Markup : It is the markup that relates to the meaning of content.

Semantic
वी Tag जिससे देखकर पता चल जाए।
meaning

`<h1>`

`<p>`

`<img>`

Non Semantic
वी Tag जिससे देखकर पता न चले।
meaning

`<div>`

`<span>`

Advantages -

- meaningful / layout → structured
- SEO friendly
- Readable + screen reader



user experience → improve.

Semantic Tags :-

<header> </header>

<main> </main>

<footer> </footer>

<nav> </nav>

<article> </article>

<section> </section>

<aside> </aside>

HTML Entities :

- An HTML entity is a piece of text ("string") that begins with an ampersand (&) and ends with a semicolon (;).
- Used to display reserved characters (which would otherwise be interpreted as HTML code), and invisible characters (like non-breaking spaces).
- Can also use in place of characters that are difficult to type with a standard keyboard.
- browser interprets them & renders correct character.

• space $\Rightarrow$ ` ` ;

$\searrow$ non breaking space.

`<` $\Rightarrow$ `<` ;

`>` $\Rightarrow$ `>` ;

`"` $\Rightarrow$ `"` ;

`&` $\Rightarrow$ `&` ;

Emmet : Short form of Code
(visit emmet.io)

child : `>`

`nav > ul > li`

`<nav>`

`<ul>`

`<li> </li>`

`</ul>`

`</nav>`

Sibling : `+`

`div + p + bq`

`<div> </div>`

`<p> </p>`

`<blockquote> </blockquote>`

Multiplication : `*`

`ul > li * 2`

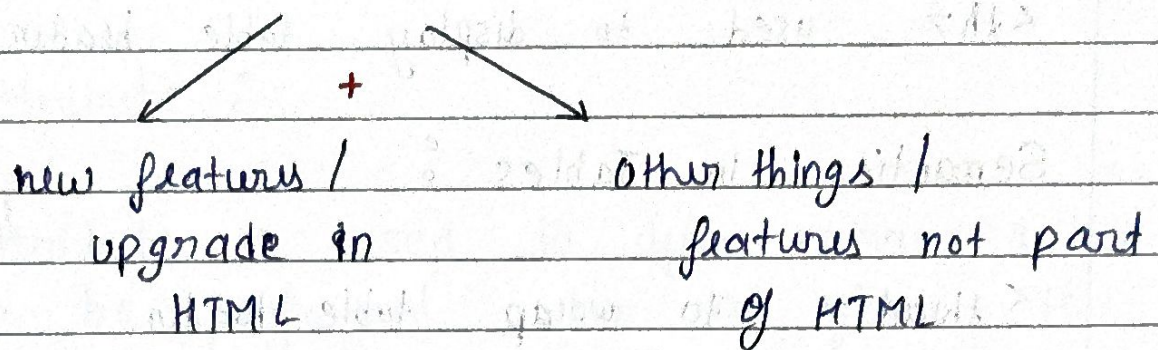
`<ul>`

`<li> </li>`

`<li> </li>`

HTML 5 : The term HTML 5 is essentially a buzzword that refers to a set of modern web technologies.

This includes the HTML Living Standard, along with JavaScript APIs to enhance storage, multimedia and hardware access.



How HTML works ?

HTML Standard : It is a document that tells to the browser how HTML should work.

10. Tables in HTML :

<table> Ⓢ <table border = "black">

<caption> Table Caption </caption>

<tr>

<th> header 1 </th>

<th> header 2 </th>

</tr>

<tr>

<td> data 1 </td>

<td> data 2 </td>

</tr>

</table>

Table Tags in HTML :

`<table>` used to display table now

`<td>` used to display table data

`<th>` used to display table header

Semantics in Tables :

`<thead>` to wrap table header

`<tbody>` to wrap table body

`<tfoot>` to wrap table footer

Colspan & Rowspan Attributes :

Used to create cells which spans over multiple rows or columns.

`rowspan` = " "
 `colspan` = " "

caption → Food Menu

Item	Price	
	INR	USD
Frooti	10	0.14
Samosa	12	0.17
Chips	20	0.28

`<th colspan="2"> Price </th>`

`<th rowspan="2"> Item </th>`

price → 1 row

2 col

Item → 2 row

1 col

FORMS in HTML : Forms are used to collect data from the user

`< form >`

form content

`< /form >`

Action Attribute :

Action attribute is used to define what action needs to be performed when a form is submitted or where the form data should be sent.

`< form action = "/action.php" >`

`< form action = "/action" >`

Form Elements :

Input

- Used to create multiple form controls.
- There are multiple types of inputs that can be created using type attribute.

`< input type = " " >`

type attributes

`< input type = "password" >`

`< input type = "text" >` `< input type = "number" >`

Placeholder attribute :

`<input type = "text" placeholder = "Enter Name" >`

Label

label element represents a caption for an item in a user interface.

`<label>`

Enter your username :

`<input type = "text" placeholder = "username" >`

`</label>`

Label → classical approach

`<label for = "username" > Enter your username : </label>`

same

`<input type = "text" id = "username" placeholder = "username" >`

Button Element :

`<button > Submit </button>`

Type attribute -

`<button type = "submit" > Submit </button>`

`<button type = "button" > do something </button>`

`<button type = "reset" > do something </button>`

Name Attribute :

Name of the form control.

Submitted with the form as part of a name / value pair.

```
<input type = "text" placeholder = "enter name"
```

```
id = "username" name = "username" />
```

↓
name की help से किसी भी website पर जा सकते हैं।

Checkbox :

```
<input type = "checkbox" name = "age" id = "age" checked />
```

```
<label for = "age" > I am 18+ </label>
```

Radio Button :

```
<input type = "radio" name = "fruit" id = "apple" value = "apple" />
```

ये बताता है की क्या select किया है।

```
<label for = "apple" > Apple </label>
```

Diffrence between Radio and Check box :

Checkbox में Group करने के बाद भी multiple check कर सकते हैं जबकि Radio में group करने के बाद केवल एक को check (select) कर सकते हैं।

Select Element :

Drop down :

```
<select name = "profession" id = "profession">
```

```
<option value = "student"> Student </option>
```

```
<option value = "dev"> Developer </option>
```

```
</select>
```

Range :

```
<label for = "volume"> Volume </label>
```

```
<input type = "range" min = "0" max = "100" name = "vol">
```

Text Area :

```
<label for = "feedback"> Please Provide your valuable  
feedback : </label>
```

```
<textarea id = "feedback"> </textarea>
```